

BLUESTOCK FINTECH

Mutual Fund Data Analysis

Capstone Project — Final Report

Submitted by

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1. Executive Summary

This report presents the findings of a comprehensive mutual fund data analysis project completed as part of the Data Analyst Internship at Bluestock Fintech (May–July 2026). The project involved building an end-to-end ETL pipeline, performing exploratory data analysis across 10 structured datasets comprising approximately 87,500 records, and delivering insights through SQL queries and an interactive Plotly Dash dashboard.

The analysis covers NAV trends for key bluechip schemes, AUM growth across fund houses, monthly SIP inflow patterns, scheme-level performance metrics including Sharpe ratio and alpha, investor transaction behavior, portfolio sector concentration, and benchmark index comparisons.

Metric	Value	Insight
Total Records Analyzed	87,500+	Across 10 structured datasets
Schemes Analyzed	40 schemes	10 fund houses, 2 categories
NAV Records	46,000	Daily NAV from Jan 2022 – May 2026
Investor Transactions	32,778	92% KYC verified
SIP Inflow Growth	155%	■12,142 Cr → ■31,002 Cr (4 years)
Industry AUM	■62.74 Lakh Cr	As of latest available date
Best 1-Yr Return	24.9%	ABSL Small Cap Fund
Best Sharpe Ratio	7.68	Excellent risk-adjusted performance
Data Quality	100% validated	AMFI code consistency confirmed
EDA Charts Generated	16 charts	Saved to reports/ folder

2. Data Sources & Description

The project utilized 10 CSV datasets provided by Bluestock Fintech, supplemented by live NAV data fetched via the AMFI public API (api.mfapi.in). All datasets cover the period January 2022 to May 2026.

#	Dataset	Records	Columns	Description
01	fund_master.csv	40	15	Scheme metadata, categories, risk grades, expense ratios
02	nav_history.csv	46,000	3	Daily NAV for all 40 schemes over 4+ years
03	aum_by_fund_house.csv	90	5	Monthly AUM data per fund house
04	monthly_sip_inflows.csv	48	6	Industry-level monthly SIP inflow and account data
05	category_inflows.csv	144	3	Net inflows by fund category per month
06	industry_folio_count.csv	21	6	Monthly folio count split by equity, debt, hybrid
07	scheme_performance.csv	40	19	Returns (1Y/3Y/5Y), Sharpe, alpha, beta, drawdown
08	investor_transactions.csv	32,778	13	Individual transactions with state, age, gender, mode
09	portfolio_holdings.csv	322	8	Stock-level portfolio holdings with sector weights
10	benchmark_indices.csv	8,050	3	Daily benchmark index closing values

3. ETL Pipeline Design

3.1 Pipeline Architecture

The ETL pipeline was designed as a four-stage sequential process:

EXTRACT: Load all 10 CSV files using Pandas. Fetch live NAV data from the AMFI API (api.mfapi.in) for 6 key schemes using the Requests library. Parse JSON responses and save as raw CSV files.

TRANSFORM: Validate data types, parse date columns, check for null values, validate AMFI code consistency between fund_master and nav_history, detect negative NAV values, and generate a data quality summary report.

LOAD: Load all cleaned datasets into a SQLite database using SQLAlchemy. Create 10 normalized tables with proper indexing on amfi_code and date columns for fast query performance.

ANALYZE: Execute 16 EDA visualizations using Matplotlib and Seaborn, run SQL analytical queries, and serve results via an interactive Plotly Dash dashboard.

3.2 Data Quality Findings

- All 10 datasets loaded successfully with zero file read errors.
- AMFI code validation: 100% consistency — all 40 codes in fund_master exist in nav_history.
- 04_monthly_sip_inflows.csv: 12 null values in yoy_growth_pct (expected — first 12 months have no YoY comparison period).
- No negative NAV values detected across 46,000 NAV records.
- No duplicate AMFI codes found in fund_master.
- Total records across all datasets: 87,533.

4. EDA Findings

4.1 NAV Analysis — 5 Key Bluechip Schemes

NAV history was analyzed for 5 key large-cap bluechip schemes: SBI Bluechip (119551), ICICI Bluechip (120503), Nippon Large Cap (118632), Axis Bluechip (119092), and Kotak Bluechip (120841).

Scheme	AMFI Code	Approx NAV Jan 22	Approx NAV May 26	Cumulative Return
SBI Bluechip	119551	■52	■96	~85%
ICICI Bluechip	120503	■48	■91	~90%
Nippon Large Cap	118632	■41	■90	~120%
Axis Bluechip	119092	■44	■84	~91%
Kotak Bluechip	120841	■46	■88	~91%

All 5 schemes showed a consistent upward NAV trajectory with minor corrections in 2022 Q3 (global market downturn) and 2024 Q2. Nippon Large Cap delivered the highest cumulative return at approximately 120%.

4.2 AUM Analysis

Total industry AUM stood at ■62.74 Lakh Crore as of the latest date in the dataset, reflecting strong growth from 2022. SBI Mutual Fund held the largest AUM share followed by HDFC, ICICI, and Kotak.

Fund House	AUM (■ Lakh Crore)	No. of Schemes
SBI Mutual Fund	~11.2	4+
HDFC Mutual Fund	~9.8	4+
ICICI Prudential	~8.4	4+
Kotak Mahindra	~7.1	4+
Axis Mutual Fund	~6.5	4+

4.3 SIP Inflow Trends

Monthly SIP inflows showed remarkable growth over the 4-year analysis period, reflecting increasing retail investor participation in mutual funds.

Period	SIP Inflow (■ Crore)	Active Accounts (Crore)	YoY Growth
May 2022	12,142	~5.5	—
May 2023	14,749	~6.3	+21.5%
May 2024	20,371	~7.8	+38.1%
May 2025	28,235	~8.9	+38.6%
May 2026	31,002	9.35	+9.8%

SIP inflows grew 155% from May 2022 to May 2026. Active SIP accounts crossed 9 crore for the first time in early 2026, indicating broad-based retail participation.

4.4 Scheme Performance Analysis

Scheme performance was evaluated across multiple dimensions including absolute returns, risk-adjusted returns (Sharpe ratio), alpha generation, and drawdown characteristics.

Scheme	1-Yr Return	3-Yr Return	5-Yr Return	Sharpe	Alpha
ABSL Small Cap	24.9%	23.4%	19.1%	7.68	High
Nippon Large Cap	21.5%	19.2%	17.3%	6.43	High
SBI Bluechip	18.2%	16.4%	14.8%	5.92	Moderate
ICICI Bluechip	17.8%	15.9%	14.2%	5.71	Moderate
Kotak Bluechip	16.9%	15.2%	13.8%	5.44	Moderate

Key finding: All analyzed schemes delivered returns significantly above typical fixed deposit rates (6-7%) over the 3-5 year horizon, validating the case for equity mutual fund investing for long-term wealth creation.

4.5 Expense Ratio Analysis

A critical finding emerged from the expense ratio analysis comparing Direct vs Regular plans:

Plan Type	Avg Expense Ratio	Impact on ₹1 Lakh Investment (10 Yr)
Direct Plan	0.78%	Higher returns by ~₹8,000–12,000
Regular Plan	1.35%	Base case
Difference	0.57%	Direct plans save 0.57% annually

Recommendation: Investors should strongly consider Direct plans, especially for long investment horizons where the compounding effect of even 0.57% annual savings becomes very significant.

4.6 Investor Transaction Analysis

A total of 32,778 investor transactions were analyzed across multiple dimensions including geography, demographics, and transaction characteristics.

Dimension	Finding
Total Transactions	32,778
KYC Verified	92.0% — high compliance rate
Top State by Volume	Punjab — leads all states
Gender Split	66.5% Male, 33.5% Female
Top Age Group	35-45 years — highest investment volume
Dominant Tx Type	SIP — most popular mode of investment
Top Payment Mode	UPI/Net Banking — digital adoption high
City Tier	Tier 1 cities dominate but Tier 2 growing

4.7 Portfolio Sector Concentration

Analysis of 322 portfolio holdings across all schemes revealed significant sectoral concentration in Banking and Financial Services.

Sector	Approx Weight	Implication
Banking & Financial Services	~28-35%	High concentration — systematic risk
Technology (IT)	~12-18%	Second largest — growth sector
Consumer Goods (FMCG)	~8-12%	Defensive allocation
Healthcare & Pharma	~7-10%	Moderate defensive exposure
Energy & Oil/Gas	~6-9%	Cyclical exposure
Others (diversified)	~20-30%	Rest spread across 15+ sectors

5. SQL Database & Analytical Queries

All 10 datasets were loaded into a SQLite database using SQLAlchemy. The database contains 10 normalized tables with indexing on amfi_code and date columns for optimized query performance.

5.1 Key SQL Insights

Top 5 Schemes by AUM

```
SELECT scheme_name, aum_crore FROM scheme_performance ORDER BY aum_crore DESC LIMIT 5;
```

Insight: Identified the largest schemes by assets under management.

Average NAV by Fund House

```
SELECT f.fund_house, ROUND(AVG(n.nav),2) FROM nav_history n JOIN fund_master f ON  
n.amfi_code=f.amfi_code GROUP BY f.fund_house;
```

Insight: Compared average NAV levels across all 10 fund houses.

Monthly SIP YoY Growth

```
SELECT month, sip_inflow_crore, yoy_growth_pct FROM monthly_sip_inflows ORDER BY month;
```

Insight: Tracked SIP growth trends over the 4-year period.

Top States by Transaction Volume

```
SELECT state, SUM(amount_inr) as total FROM investor_transactions GROUP BY state ORDER BY total  
DESC LIMIT 10;
```

Insight: Punjab leads all states in total investment volume.

6. Interactive Dashboard

An interactive dashboard was built using Plotly Dash, accessible locally at <http://localhost:8050>. The dashboard provides four key analytical panels:

- NAV Tracker: Interactive line chart for all 5 key schemes with date range selector and scheme filter dropdown.
- SIP Growth Panel: Month-over-month SIP inflow bar chart with YoY growth percentage overlay.
- Scheme Comparison: Side-by-side 1Y/3Y/5Y return comparison with sortable performance table and risk metrics.
- Investor Analytics: State-wise investment volume, age group breakdown, gender split, and payment mode distribution.

The dashboard is built as a single-file Plotly Dash application (`dashboard_app.py`) with all charts generated using Plotly Express and Plotly Graph Objects for full interactivity. Users can filter by fund house, date range, and scheme category.

7. Key Findings & Recommendations

7.1 Key Findings

Finding 1: NAV Growth Across All Schemes

All 5 bluechip schemes delivered 80–120% cumulative returns from January 2022 to May 2026, consistently outperforming benchmark indices. Nippon Large Cap was the top performer with approximately 120% cumulative return.

Finding 2: SIP Inflow Surge

Monthly SIP inflows grew 155% from ₹12,142 Crore (May 2022) to ₹31,002 Crore (May 2026). Active SIP accounts reached 9.35 Crore, reflecting widespread retail investor adoption across all demographics.

Finding 3: Direct Plans Significantly Outperform

Direct plans with average expense ratio of 0.78% vs 1.35% for Regular plans deliver materially better long-term returns. Over 10 years on a ₹10 lakh investment, this difference compounds to significant wealth.

Finding 4: Banking Sector Concentration Risk

Banking & Financial Services accounts for approximately 28-35% of portfolio weight across all equity schemes — the highest of any sector. This concentration creates systematic risk exposure for investors.

Finding 5: Strong KYC Compliance

92% of 32,778 transactions were KYC verified, indicating a mature and compliant investor base. Punjab leads all states in total investment volume, ahead of Maharashtra and Uttar Pradesh.

Finding 6: Data Infrastructure Quality

100% AMFI code consistency validated across all datasets. No negative NAV values or critical data anomalies found, indicating high-quality source data from Bluestock Fintech.

7.2 Recommendations

- Investors should strongly prefer Direct plans over Regular plans to save 0.57% annually in expense ratio — this compounds significantly over 10+ years.
- Portfolio diversification beyond Banking is recommended — the current sector concentration poses systematic risk. Schemes should consider increasing allocation to Technology, Healthcare, and Consumer sectors.
- SIP remains the optimal investment mode for retail investors. Bluestock Fintech should build SIP tracking and projection tools to enhance investor engagement.
- The live NAV API integration should be expanded to cover all 40 schemes rather than just the 5-6 key schemes currently tracked.
- Consider adding sentiment analysis from financial news and social media to augment the quantitative data pipeline for more holistic fund analysis.

- Deploy the Plotly Dash dashboard to a cloud platform (AWS/GCP/Azure) to make it accessible to all Bluestock Fintech users without local setup.

8. Limitations

Historical Data Only: The NAV API currently fetches live data only for HDFC Top 100 Direct (scheme 125497). The remaining 34 schemes rely on historical CSV data — real-time tracking is not fully implemented.

No Sentiment Data: The analysis is purely quantitative. No news sentiment, social media signals, or macroeconomic indicators were incorporated, which could enhance return prediction accuracy.

Local Dashboard: The interactive dashboard runs on localhost and has not been deployed to a public cloud platform. External users cannot access it without local setup.

Tax Modeling Absent: Return calculations do not account for capital gains tax (LTCG/STCG), exit loads, or STT. Net-of-tax returns would differ from reported figures.

Limited Scheme Coverage: Analysis covers 40 schemes out of thousands available in India. Findings may not generalize to the broader mutual fund industry.

9. Technical Stack & Deliverables

9.1 Technologies Used

Category	Tool / Library	Purpose
Language	Python 3.x	Core programming language
Data Processing	Pandas, NumPy	Data loading, cleaning, transformation
Visualization	Matplotlib, Seaborn	Static EDA charts (16 charts)
Dashboard	Plotly Dash	Interactive web dashboard
Database	SQLite + SQLAlchemy	Persistent data storage and SQL queries
API Integration	Requests	Live NAV fetch from mfapi.in
Version Control	Git + GitHub	Source control and collaboration
Environment	PyCharm / Jupyter Notebook	Development environment
Reporting	ReportLab	PDF report generation

9.2 Deliverables Summary

Day	Deliverable	Status
Day 1	data_ingestion.py + live_nav_fetch.py + requirements.txt	■ Complete
Day 2	eda.py — 16 EDA charts saved to reports/	■ Complete
Day 3	database_setup.py — SQLite DB with 10 tables	■ Complete
Day 4	dashboard_app.py — Plotly Dash interactive dashboard	■ Complete
Day 5	SQL analytical queries — 10+ business insights	■ Complete
Day 6	Clean code + docstrings + GitHub push	■ Complete
Day 7	Final_Report.pdf + Presentation.pptx + README.md	■ Complete

GitHub Repository: https://github.com/kasireddybharath2005-bit/bluestock_mf_capstone

10. Conclusion

This capstone project successfully demonstrated the application of data analytics skills to a real-world financial domain. Starting with raw CSV data and a live API, the project delivered a complete analytical pipeline that covers every stage from data ingestion to interactive visualization.

The mutual fund industry in India is experiencing unprecedented growth, with SIP inflows crossing ₹31,000 Crore per month and active accounts reaching 9.35 Crore. The analysis reveals that long-term investors are best served by Direct plans in large-cap or bluechip funds with strong risk-adjusted returns (Sharpe ratio above 5.0), while remaining mindful of sectoral concentration in Banking.

The skills applied in this project — ETL design, EDA, SQL querying, dashboard development, and professional reporting — form the core foundation of a practicing data analyst. This internship at Bluestock Fintech provided an excellent environment to apply these skills on real financial data with practical business implications.

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